Experiment 4.1.1

Algorithm:

Algorithm: Set Operations on Set A and Set B

Step 1. Start

Step 2.Read elements of Set A ans Set B

Step 3. Compute Union
Union = Set_A ∪ Set_B

Step 4. Compute Intersection
Intersection = Set_A ∩ Set_B

Step 5.Compute Difference
Difference = Set_A − Set_B

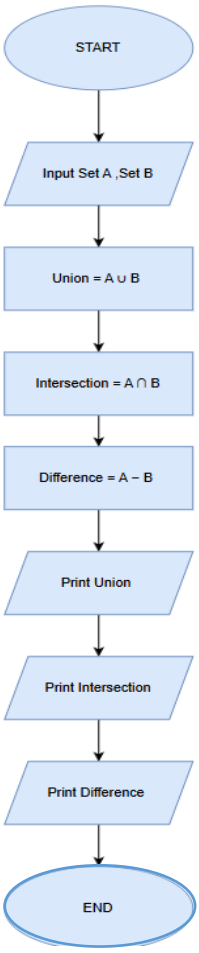
Step 6. Print Union

Step 7. Print Intersection

Step 8. Print Difference

Step 9. Stop

Flowchart:



Course

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4.1.1. Set Operations

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Write a Python program to perform union, intersection and difference operations on $Set\ A$ and $Set\ B$.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for $Set\ A$.
- The second input prompts "Set B: " followed by space-separated list of integers for $Set\ B$.

Output Format:

- The first line prints "Union: " followed by the union of $Set\ A$ and $Set\ B$.
- The second line prints "Intersection: " followed by the intersection of $Set\ A$ and $Set\ B$.
- The third line prints "Difference: " followed by the difference of $Set\ A$ and $Set\ B$.

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases +

Explorer setoperat... Submit Debugger

```python
Set_a = set(map(int, input("Set A: ").split()))
Set_b = set(map(int, input("Set B: ").split()))

union = Set_a | Set_b
intersection = Set_a & Set_b
difference = Set_a - Set_b

print("Union:", union)
print("Intersection:", intersection)
print("Difference:", difference)
```

Average time	Maximum time
0.007 s	0.010 s
7.00 ms	10.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output	Actual output
Set A: 0 2 4 5 8	Set A: 0 2 4 5 8
Set B: 1 2 3 4 5	Set B: 1 2 3 4 5
Union: {0, 1, 2, 3, 4, 5, 8}	Union: {0, 1, 2, 3, 4, 5, 8}
Intersection: {2, 4, 5}	Intersection: {2, 4, 5}
Difference: {0, 8}	Difference: {0, 8}

Terminal Test cases

‹ Prev Reset Submit Next ›

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